

The University of Nottingham

SCHOOL OF MATHEMATICAL SCIENCES

A LEVEL 3 MODULE, SPRING SEMESTER 2023-2024

TIME SERIES ANALYSIS

Time allowed THREE Hours

Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced.

Answer ALL questions

Only a calculator from approved list A may be used in this examination.

List A

Basic Models	Scientific Calculators
Aurora HC133	Aurora AX-582
Casio HS-5D	Casio FX82 family
Deli – DL1654	Casio FX83 family
Sharp EL-233	Casio FX85 family
	Casio FX350 family
	Casio FX570 family
	Casio FX 991 family
	Sharp EL-531 family
	Texas Instruments TI-30 family
	Texas BA II+ family

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

1. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2 .

The process $\{X_t : t \in \mathbb{Z}\}$ is such that

$$X_t = Z_t - \frac{11}{12}Z_{t-1} + \frac{1}{6}Z_{t-2}.$$

- (a) Show that this process is invertible. [4 marks]
- (b) Determine the autocovariance generating function of $\{X_t : t \in \mathbb{Z}\}$. [3 marks]
- (c) Hence write down the autocovariance function of $\{X_t : t \in \mathbb{Z}\}$ in terms of σ_z^2 . Explain your answer. [4 marks]
- (d) Determine the spectral density function of $\{X_t : t \in \mathbb{Z}\}$, expressing your answer in terms of $\cos w$ (where $w \in (-\pi, \pi]$). [4 marks]
- (e) Sketch the spectral density function from (d) for frequency values in the range $w \in [0, \pi]$, justifying the shape of your sketch. What does this spectral density function suggest with regard to the likely behaviour of the process $\{X_t : t \in \mathbb{Z}\}$? Explain your answer. [10 marks]

2. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2 .

The (weakly) stationary process $\{X_t : t \in \mathbb{Z}\}$ is such that

$$X_t = \mu - \frac{1}{9}(X_{t-1} - \mu) + \frac{2}{27}(X_{t-2} - \mu) + Z_t$$

where μ is a real-valued parameter and $\mathbb{E}(X_t) = \mu$ for all $t \in \mathbb{Z}$.

- (a) The process $\{Y_t : t \in \mathbb{Z}\}$ is such that $Y_t = X_t - \mu$. Show that

$$\gamma_Y(h) = \gamma_X(h)$$

where $\gamma_X(h)$ and $\gamma_Y(h)$ are the autocovariance functions of $\{X_t : t \in \mathbb{Z}\}$ and $\{Y_t : t \in \mathbb{Z}\}$, respectively, with $h \in \mathbb{Z}$.

Hint: You do not need to determine $\gamma_X(h)$ or $\gamma_Y(h)$ explicitly here.

[2 marks]

- (b) For the process $\{Y_t : t \in \mathbb{Z}\}$, derive the Yule-Walker equations. You may assume that Y_{t-h} and Z_t are independent where $h > 0$ is a positive integer.

[5 marks]

- (c) By solving the Yule-Walker equations from (b), determine the autocorrelation function for the process $\{Y_t : t \in \mathbb{Z}\}$. You should show your working clearly.

[7 marks]

- (d) Show that $\mathbb{E}(Y_t Z_t) = \sigma_z^2$.

[4 marks]

- (e) A sequential sample of $\{X_t : t \in \mathbb{Z}\}$ (of size n) is observed, denoted $x_1, \dots, x_n$. Show that an approximate estimate of σ_z^2 is

$$\hat{\sigma}_z^2 = c_0 \left(1 + \frac{1}{9}r_1 - \frac{2}{27}r_2 \right)$$

where $c_0 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ and r_j is the sample autocorrelation of the process $\{X_t : t \in \mathbb{Z}\}$ at lag j ($j \in \{1, 2\}$). Supposing $c_0 = 0.94$, $r_1 = -0.23$ and $r_2 = 0.15$, estimate σ_z^2 .

[7 marks]

3. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2 .

The process $\{X_t : t \in \mathbb{Z}\}$ is such that

$$X_t = 0.5X_{t-1} + Z_t - 0.4Z_{t-1}.$$

- (a) What type of process is $\{X_t : t \in \mathbb{Z}\}$, including its order?

[1 mark]

- (b) Is the process $\{X_t : t \in \mathbb{Z}\}$ (weakly) stationary? Justify your answer.

[3 marks]

- (c) Write the process $\{X_t : t \in \mathbb{Z}\}$ as a moving average process of infinite order (i.e. in $MA(\infty)$ form). You should show your working clearly and express your final answer in its most simplified form.

[8 marks]

- (d) Using your answer to (c), determine the autocorrelation function of $\{X_t : t \in \mathbb{Z}\}$.

[9 marks]

- (e) A large sequential sample of $\{X_t : t \in \mathbb{Z}\}$ is taken but the following model is fitted to the observed data:

$$X_t = 0.5X_{t-1} + Z_t.$$

The residuals from this fitted model are calculated and a plot of the sample autocorrelation function of these residuals is produced for lags 0 to 15. Describe how you would expect this plot to look. Explain your answer.

[4 marks]

4. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2 .

The process $\{X_t : t \in \mathbb{Z}\}$ is an MA(1) process, written

$$X_t = Z_t + \theta Z_{t-1}$$

with $|\theta| < 1$.

- (a) Determine the autocorrelation function of $\{X_t : t \in \mathbb{Z}\}$. You should show your working clearly.

[4 marks]

- (b) A sequential sample of $\{X_t : t \in \mathbb{Z}\}$ of size n is recorded, denoted $x_1, \dots, x_n$. Suppose that $\bar{x}$ is the sample mean, $c_0 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ and r_1 is the sample autocorrelation at lag 1. Determine method of moments estimates of θ and σ_z^2 in terms of c_0 and r_1 .

[5 marks]

- (c) Write the process $\{Z_t : t \in \mathbb{Z}\}$ in the form

$$Z_t = \sum_{j=0}^{\infty} a_j X_{t-j}$$

where the coefficients a_j are real-valued functions of θ . You should show your working clearly.

[5 marks]

- (d) The process $\{Y_t : t \in \mathbb{Z}\}$ measures the annual rainfall in millimetres (mm) at a weather station and it is assumed that

$$Y_t = \mu + Z_t + \psi Z_{t-1}$$

where μ and ψ are real-valued parameters. Suppose that we have 30 years of observations of this process from 1994 to 2023, denoted $y_{1994}, \dots, y_{2023}$. The sample mean of these observations is $\bar{y} = 705.6$ and the sample autocorrelation at lag 1 is $r_1 = 0.48$. In addition, $\frac{1}{30} \sum_{i=1994}^{2023} (y_i - \bar{y})^2 = 192.1$.

Calculate the method of moments estimates of the parameters μ , ψ and σ_z^2 .

[4 marks]

- (e) The observed values of the process $\{Y_t : t \in \mathbb{Z}\}$ (described in (d)) for the years 2019–2023 are shown in the table below.

Year	2019	2020	2021	2022	2023
Annual rainfall (mm)	705	722	685	682	701

Using the above information, together with your parameter estimates from (d), provide an optimal prediction for the 2024 annual rainfall. Here, the term ‘optimal’ implies that the predicted value is that which minimises the mean-square prediction error at a given time point. Note: If you did not obtain an answer to (d) you may assume that $\hat{\mu} = 704$, $\hat{\psi} = 0.8$ and $\hat{\sigma}_z^2 = 120$ (these are not the correct answers to (d)).

[7 marks]